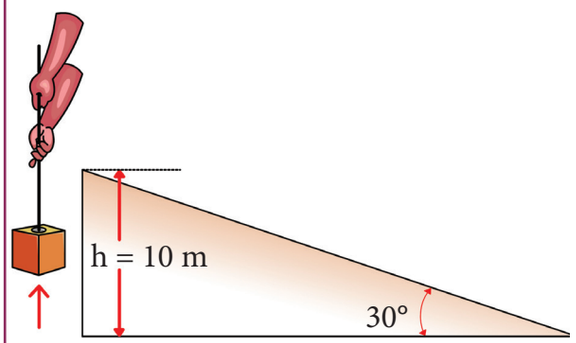
$$E = KE = \frac{1}{2}mv^2 = 100 \text{ J}$$

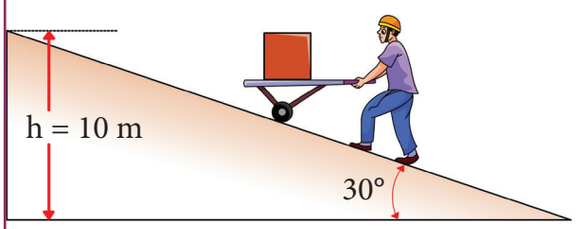
$$v = \sqrt{\frac{2}{m}KE} = \sqrt{\frac{2}{1} \times 100} = \sqrt{200} \approx 14.12 \text{ m s}^{-1}$$

EXAMPLE 4.14

A body of mass 100 kg is lifted to a height 10 m from the ground in two different ways as shown in the figure. What is the work done by the gravity in both the cases? Why is it easier to take the object through a ramp?



Path (1) straight up



Path (2) along the ramp

Solution

m = 100 kg, h = 10 m

Along path (1):

The minimum force F_1 required to move the object to the height of 10 m should be equal to the gravitational force, $F_1 = mg = 100 \times 10 = 1000$ N

The distance moved along path (1) is, $h = 10$ m

The work done on the object along path (1) is

$$W = F\ h = 1000 \times 10 = 10{,}000 \text{ J}$$

Along path (2):

In the case of the ramp, the minimum force F_2 that we apply on the object to take it up is not equal to mg, it is rather equal to $mg \sin\theta$. $(\text{mg}\sin\theta < mg)$.

Here, angle θ = 30°

Therefore, F_2 = mg sinθ = 100 × 10 × sin30° = 100 × 10 × 0.5 = 500 N

Hence, $(\text{mg}\sin\theta < mg)$

The path covered along the ramp is, $l = \frac{h}{\sin 30°} = \frac{10}{0.5} = 20m$

The work done on the object along path (2) is, $W = F_2 l = 500 \times 20 = 10{,}000$ J

Since the gravitational force is a conservative force, the work done by gravity on the object is independent of the path taken.

In both the paths the work done by the gravitational force is 10,000 J

Along path (1): more force needs to be applied against gravity to cover lesser distance .

Along path (2): lesser force needs to be applied against the gravity to cover more distance.

As the force needs to be applied along the ramp is less, it is easier to move the object along the ramp.

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